

MHF 4U - Chapter 1 Polynomial Functions Test

Name _____

Date _____

Total
27.5
/31

KNOWLEDGE Identify the choice that best completes the statement or answers the question.

B 1. The function $f(x) = 6(x-1)^4(x-2)^2(x+1)$ changes signs at

- a. $x = -2$ b. $x = -1$ c. $x = 2$ d. does not change signs



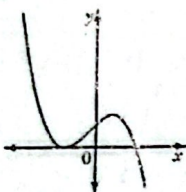
B 2. The graph of the polynomial function $f(x) = -2(x+1)^2(-3x+1)^3$ has a leading coefficient of

- a. -6 b. -2 c. 54 d. 6

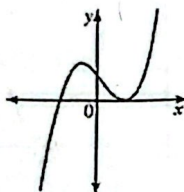
$$-2(1)(-3)^3 = 54$$

A 3. Which graph most closely resembles the function $f(x) = -2(x-1)(x+1)^2$?

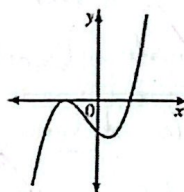
a)



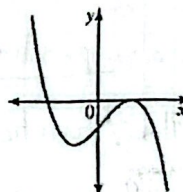
b)



c)



d)

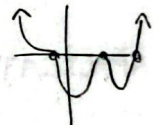


C 4. What is the minimum number of local max/min points that $f(x) = -x^6 + 12x^5 - 8x - 1$ can have?

- a. 0 b. 2 c. 1 d. 3

D 5. The function $f(x) = (x-4)^2(x-7)(x+3)^3$ is negative on the interval(s)

- a. $(-\infty, -3)$ and $(4, 7)$ b. $(-3, 4)$ and $(7, \infty)$ c. $(-\infty, 3)$ and $(7, \infty)$ d. $(-3, 4)$ and $(4, 7)$



B 6. Given the function, $f(x) = -2(3x+6)^4 + 1$, which of the following statements is **not true** when compared to the parent function $y = x^4$?

- a. Vertical stretch by a factor of 2 b. Reflection on the y-axis
c. Reflection on the x-axis d. Horizontal translation two units left

7. Determine the equation of a **sixth degree** polynomial function with the following characteristics:

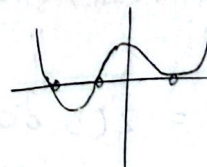
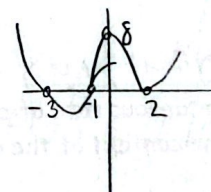
- zeros at -3, -1 and 2
- is **ONLY** positive on the intervals $(-\infty, -3)$, $(-1, 2)$ and $(2, \infty)$
- y-intercept of (0, 8)

$$f(x) = a(x+3)(x+1)(x-2)^4$$

$$8 = a(0+3)(0+1)(0-2)^4$$

$$a = \frac{1}{6}$$

$$f(x) = \frac{1}{6}(x+3)(x+1)(x-2)^4$$



(6)

(3)

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COMMUNICATION

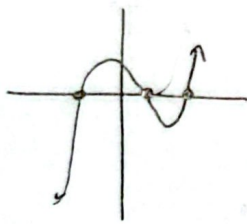
1. Are all even degree polynomial functions even? Provide an example to support your reasoning.

no, in order to be an even function, the even degree polynomial must fulfill $f(x) = f(-x)$ and have line symmetry about the y-axis - not all even degree polynomials fulfill these. (2)

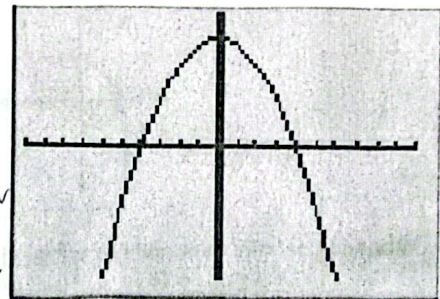
for example, $f(x) = (x-3)^4(x-7)^2$

$f(-x) = (-x-3)^4(-x-7)^2 \rightarrow (x+3)^4(x+7)^2$] $f(x) \neq f(-x)$ and roots are not the same

2. This graph below represents a portion of the function $f(x) = x^3 - 250x^2 + 4000$. The function appears to have some two zeros. A third zero exists for this function. Is the third zero a positive or negative value? Explain.



the third 0 must be a positive value because the function is degree 3 with a pos. LC so in order to fulfill Q3 to Q1, end, behaviour, a third 0 must exist bringing the function from Q4 to Q1 as right now the function on the graph ends at Q4



APPLICATION

1. The amount of money, M in dollars, in a piggy bank after t days is given by the formula $M(t) = 2t^3 + 2t + 10$

- a. Determine the average rate of change of the amount of money in the piggy bank over the interval $[1, 4]$.

$$2t^3 + 2t + 10$$

$$f(1) = 2(1)^3 + 2(1) + 10 = 14$$

$$f(4) = 2(4)^3 + 2(4) + 10 = 146$$

$$AROC = \frac{146 - 14}{4 - 1} = \frac{132}{3} = 44$$

the AROC is \$44/day (2)

- b. Estimate the instantaneous rate of change in the amount of money in the piggy bank at 3 days. Interpret the solution for the context of the question.

$$f(3) = 2(3)^3 + 2(3) + 10 = 70$$

$$IRIC = \frac{70.000056 - 70}{3.000001 - 3} = 56$$

(2)

$$f(3.000001) = 2(3.000001)^3 + 2(3.000001) + 10 = 70.000056$$

the instantaneous rate of change at 3 days is \$56/day, this means at exactly 3 days, there is \$56 added (the rate of change for day 3)

$$f(x) = \frac{1}{2} \left[-\frac{1}{3}(x+5) \right]^3 + 3$$

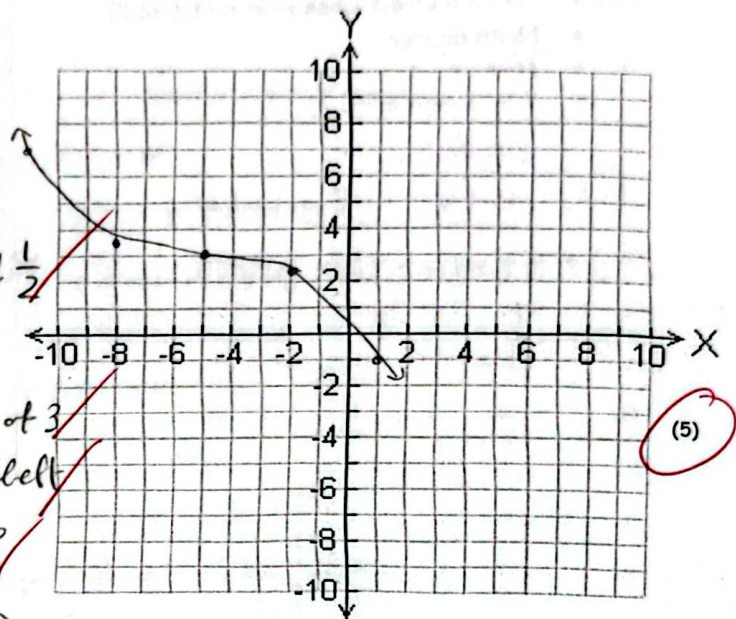
2. Sketch the graph of the function $f(x) = \frac{1}{2} \left(-\frac{1}{3}x - \frac{5}{3} \right)^3 + 3$. State the parameters of the transformed function and describe each transformation. State the mapping rule. Provide 5 points. Show all your work.

Mapping Rule: $(-3x-5, \frac{1}{2}y+3)$

$(x, y) \rightarrow (-3x-5, \frac{1}{2}y+3)$

transformations

- vertical stretch by a factor of $\frac{1}{2}$
- reflection across y-axis
- horizontal stretch by a factor of 3
- horizontal translation 5 units left
- vertical translation 3 units up



$f(x) = x^3$

x	y
-2	-8
-1	-1
0	0
1	1
2	8

$(-3x-5, \frac{1}{2}y+3)$

x	y
-1	-1
-2	2.5
-5	3
-8	3.5
-11	7

3. Algebraically determine whether the function $f(x) = -3x^4(x-1)(x+1)$ is odd, even or neither. Show all work to disprove the other two options.

$f(x) = -3x^4(x-1)(x+1)$

roots = 0, -1, 1

$$f(-x) = -3(-x)^4(-x-1)(-x+1)$$

$$= -3(x)^4(-1)(x+1)(-1)(x-1)$$

$$= -3x^4(x+1)(x-1)$$

roots = 0, -1, 1

$$-f(x) = -[-3x^4(x-1)(x+1)]$$

$$= 3x^4(x-1)(x+1)$$

↑ no - LC

the function is even as
 $f(x) = f(-x)$
 check: $f(4) = -11520$
 $f(-4) = -11520$

(2)

7

THINKING

1. a) Draw a graph of the polynomial function that satisfies these characteristics:

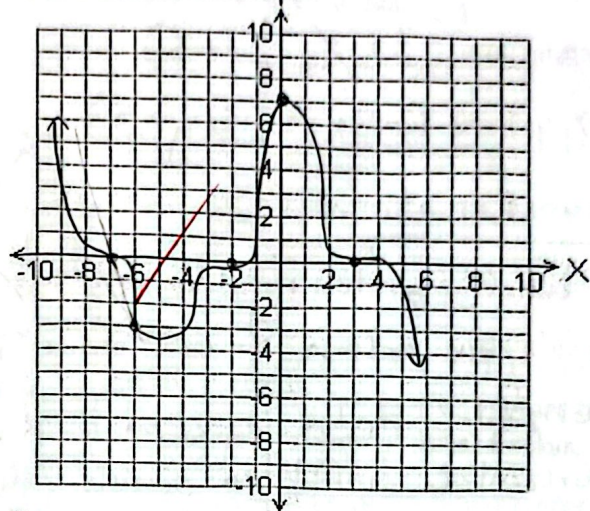
- $f(x) > 0$ ONLY when $(-\infty, -7), (-2, 3)$
- Ninth degree
- $f(-6) = -3$
- Y-intercept of $(0, 7)$

$$f(x) = a(x+7)^3(x+2)^3(x-3)^3$$

$$-3 = a(-6+7)^3(-6+2)^3(-6-3)^3$$

$$-3 = a(1)(-4)^3(-9)^3$$

$$a = -\frac{1}{16552}$$



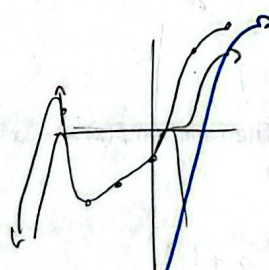
* all roots are order 3

b) Possible Equation of the Graph: $f(x) = -\frac{1}{16552}(x+7)^3(x+2)^3(x-3)^3$

2. Determine the equation of the function in standard form, $f(x) = ax^3 + bx^2 + cx + d$ given the table:

x	y
-3	4
-2	-29
-1	-26
0	-5
1	16
2	19

d value
is -5



$$f(x) = -3x^3 + bx^2 + cx + 5$$

$$16 = -3(1)^3 + b(1)^2 + c(1) + 5$$

$$24 = b + c$$

use substitution

$$FD = an!$$

$$-18 = a \cdot (3 \times 2 \times 1)$$

$$a = -3$$

$$f(x) = -3(x+s)(x+n_1)(x+n_2)$$

$$16 = -3(1+s)(1+n_1)(1+n_2)$$

$$-\frac{16}{3} = (x+n_1)(x+n_2)$$

$$-\frac{8}{9} = (x+n_1)(x+n_2)$$

$$f(x) = -3(x+s)(x+n_1)(x+n_2)$$

$$16 = -3(2+s)(2+n_1)(2+n_2)$$

$$b = 0$$

$$f(x) = 3x^3 + 24x - 5$$

4.5